

# Series of Real Numbers

## Convergence, Divergence, and Properties

August 25, 2025

# What is a Series?

## Definition

A **series** is the sum of the terms of a sequence. Given a sequence  $\{a_n\}$ , the series is denoted by:

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## Definition

A series **converges** if the sequence of partial sums  $\{S_N\}$  converges to a finite limit  $S$ :

$$\lim_{N \rightarrow \infty} S_N = S$$

# Examples of Convergence and Divergence

## Example (Geometric Series)

The geometric series  $\sum_{n=0}^{\infty} ar^n$  converges to  $\frac{a}{1-r}$  if  $|r| < 1$  and diverges if  $|r| \geq 1$ .

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## Example (Telescoping Series)

The series  $\sum_{n=1}^{\infty} \left( \frac{1}{n} - \frac{1}{n+1} \right)$  converges to 1:

$$S_N = \left( 1 - \frac{1}{2} \right) + \left( \frac{1}{2} - \frac{1}{3} \right) + \cdots + \left( \frac{1}{N} - \frac{1}{N+1} \right) = 1 - \frac{1}{N+1}$$

$$\lim_{N \rightarrow \infty} S_N = 1$$

# Necessary Condition for Convergence

## Theorem

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## Proof.

Let  $S = \lim_{N \rightarrow \infty} S_N$ . Then:

$$a_n = S_n - S_{n-1} \Rightarrow \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} (S_n - S_{n-1}) = S - S = 0$$



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## Important Note

The converse is not true! If  $\lim_{n \rightarrow \infty} a_n = 0$ , the series may still diverge.

# Counterexample: Harmonic Series

[Harmonic Series - Divergence Proof] The harmonic series  $\sum_{n=1}^{\infty} \frac{1}{n}$  satisfies  $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$  but diverges.

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Proof.

We show that the sequence of partial sums  $S_N = \sum_{n=1}^N \frac{1}{n}$  is **unbounded**. □

# Step 1: Grouping the Terms

Consider the partial sum  $S_{2^m}$  with  $N = 2^m$  terms:

$$S_{2^m} = \sum_{n=1}^{2^m} \frac{1}{n} = 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^{m-1}+1} + \cdots + \frac{1}{2^m}\right)$$

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Group the terms into blocks where the  $k$ -th block ( $k \geq 1$ ) contains the  $2^{k-1}$  terms from  $n = 2^{k-1} + 1$  to  $n = 2^k$ :

$$\text{Block } k = \sum_{n=2^{k-1}+1}^{2^k} \frac{1}{n}$$

## Step 2: Estimating Each Block

$$k = 1 : \frac{1}{2}$$

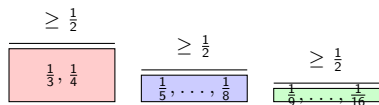
$$k = 2 : \frac{1}{3} + \frac{1}{4} > \frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$$

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$$k = 4 : \frac{1}{9} + \cdots + \frac{1}{16} > 8 \times \frac{1}{16} = \frac{1}{2}$$

$\vdots$

$$k : \sum_{n=2^{k-1}+1}^{2^k} \frac{1}{n} > 2^{k-1} \times \frac{1}{2^k} = \frac{1}{2}$$



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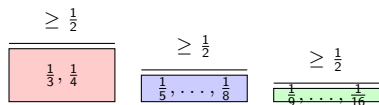
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Each block has  $2^{k-1}$  terms, each term is  $\geq \frac{1}{2^k}$ .

$$\sum_{\text{Block } k} > 2^{k-1} \times \frac{1}{2^k} = \frac{1}{2}$$



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Thus, we have shown that for any integer  $m \geq 1$ ,

$$S_{2^m} > 1 + \frac{m}{2}$$

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For any real number  $M$  (no matter how large), choose an integer  $m > 2(M - 1)$ .

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### Conclusion

Therefore, the sequence of partial sums  $\{S_N\}$  is **unbounded**. Since it is also increasing, it cannot converge to a finite limit. The harmonic series **diverges**.

# Addition of Series

## Theorem

If  $\sum a_n$  and  $\sum b_n$  are convergent series with sums  $A$  and  $B$  respectively, then:

- ①  $\sum(a_n + b_n)$  converges to  $A + B$
- ②  $\sum ca_n$  converges to  $cA$  for any constant  $c$



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## Example

$$\sum_{n=1}^{\infty} \left( \frac{1}{2^n} + \frac{1}{n(n+1)} \right) = \sum_{n=1}^{\infty} \frac{1}{2^n} + \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1 + 1 = 2$$

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## Warning

If both series diverge, their sum may converge or diverge. If one converges and one diverges, their sum always diverges.

# Case 1: One Converges, One Diverges

## Theorem

*If  $\sum a_n$  converges and  $\sum b_n$  diverges, then  $\sum(a_n + b_n)$  diverges.*

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## Proof.

Assume, for contradiction, that  $\sum(a_n + b_n)$  converges. Since  $\sum a_n$  also converges, we can use the linearity rule:

$$\sum b_n = \sum[(a_n + b_n) - a_n] = \sum(a_n + b_n) - \sum a_n$$

This would mean  $\sum b_n$  is the difference of two convergent series and must therefore converge. **Contradiction.** Therefore,  $\sum(a_n + b_n)$  must diverge. □

# Example: Convergent + Divergent = Divergent

## Example

Let  $\sum a_n = \sum_{n=1}^{\infty} \frac{1}{2^n}$  (converges, geometric series,  $r = 1/2$ ).

Let  $\sum b_n = \sum_{n=1}^{\infty} 1$  (diverges, harmonic series without decay).

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The general term does not go to zero:

$$\lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} \left( \frac{1}{2^n} + 1 \right) = 0 + 1 = 1 \neq 0$$

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By the **necessary condition for convergence** ( $n$ th term test), the series **must diverge**.



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- Converge (through **cancellation** or **telescoping**)

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Again, by the  $n$ th term test, the series **diverges**.

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### Example

A less trivial example:

Let  $\sum a_n = \sum_{n=1}^{\infty} \left( \frac{1}{n} + \frac{1}{n+1} \right)$  (diverges, like harmonic).

Let  $\sum b_n = \sum_{n=1}^{\infty} \left( -\frac{1}{n+1} - \frac{1}{n} \right)$  (diverges, same reason).

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Their sum:

$$\sum (a_n + b_n) = \sum_{n=1}^{\infty} 0 = 0 \quad (\text{converges})$$

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This is a **telescoping series**!

$$\begin{aligned} S_N &= \left( 1 - \frac{1}{2} \right) + \left( \frac{1}{2} - \frac{1}{3} \right) + \cdots + \left( \frac{1}{N-1} - \frac{1}{N} \right) \\ &= 1 - \frac{1}{N} \end{aligned}$$

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So,  $\lim_{N \rightarrow \infty} S_N = 1$ . The sum of two divergent series **converges** to 1.



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An increasing sequence converges if and only if it is bounded above.

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## Example

The series  $\sum_{n=1}^{\infty} \frac{1}{n^2}$  converges because the partial sums are bounded above (by 2, in fact).

# Introduction and Removal of Brackets

## Theorem (Introduction of Brackets)

If we group terms of a convergent series, the new series with brackets converges to the same sum.

# Introduction and Removal of Brackets

## Theorem (Introduction of Brackets)

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## Example

$$\sum_{n=1}^{\infty} (-1)^{n+1} = 1 - 1 + 1 - 1 + \cdots \quad (\text{diverges})$$

But with brackets:

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## Theorem (Removal of Brackets)

For series of positive terms, removal of brackets does not affect convergence.

# Two Types of Convergence

When a series  $\sum a_n$  converges, we can classify it further based on the behavior of its absolute value.

## Absolute Convergence

A series  $\sum a_n$  **converges absolutely** if the series of its absolute values  $\sum |a_n|$  also converges.

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## Conditional Convergence

A series  $\sum a_n$  **converges conditionally** if it converges but the series of its absolute values  $\sum |a_n|$  diverges.

$$\text{Convergence: } \sum a_n \quad \text{but} \quad \text{Divergence: } \sum |a_n|$$

# The Prime Example: Alternating Harmonic Series

## Example

The **Alternating Harmonic Series** is the classic example of conditional convergence.

$$\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \dots$$

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## Therefore...

The Alternating Harmonic Series **converges conditionally**.

# Rearrangements of Terms

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## Example

The alternating harmonic series  $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$  converges conditionally to  $\ln 2$ . But we can rearrange it to converge to any real number, or even diverge!

# Summary

- Series represent infinite sums of sequences
- Convergence depends on the behavior of partial sums
- $\lim a_n = 0$  is necessary but not sufficient for convergence
- Series of positive terms have special convergence properties
- Brackets and rearrangements can affect convergence for conditionally convergent series
- Absolute convergence guarantees stability under rearrangements